

Disease	Enzyme / Protein Defect	Key Features
1. Hyperammonemia Type I	Carbamoyl phosphate synthetase I (CPS-I) deficiency	• Very high NH₃ (ammonia) levels in blood • Autosomal recessive • Intellectual disability / developmental impairment • Incidence: ~1 in 100,000
2. Hyperammonemia Type II	Ornithine transcarbamylase (OTC) deficiency	• High ammonia in blood • Increased glutamine in blood, CSF & urine • Orotic aciduria due to diversion of carbamoyl phosphate into pyrimidine synthesis • X-linked
3. Hyperornithinemia–Hyperammonemia–Homocitrullinuria (HHH) Syndrome	Defective ornithine transporter protein (ORNT1)	• Failure to transport ornithine from cytoplasm into mitochondria • ORNT1 gene defect • Hyperornithinemia + hyperammonemia + homocitrullinuria • Decreased blood urea • Autosomal recessive
4. Citrullinemia Type I	Argininosuccinate synthetase (ASS) deficiency	• Autosomal recessive • High blood ammonia & citrulline • Citrullinuria (≈1–2 g/day)
5. Citrullinemia Type II	Defective Citrin protein (<i>Aspartate–glutamate transporter</i>)	• Decreased availability of aspartate for the argininosuccinate synthetase reaction • Fatty liver

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		• Intrahepatic cholestasis • Increased citrulline
6. Argininosuccinic Aciduria	Argininosuccinate lyase (ASL) deficiency	• Increased argininosuccinate in blood & urine • Friable, brittle, tufted hair • Trichorrhexis nodosa • Incidence: ~3 in 200,000
7. Hyperargininemia	Arginase deficiency	• Increased arginine in blood & CSF • Cysteine & lysine lost in urine instead of arginine • Incidence: ~1 in 100,000